

**INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG****B.TECH SYLLABUS for COMPUTER SCIENCE &ENGINEERING**

(Admission Batch: 2018-19 Onwards)

**4<sup>th</sup> Semester**

| <b>HSHM3203</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ENGINEERING ECONOMICS</b> | <b>3-0-0</b> | <b>Credit-3</b> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------|-----------------|
| <p><b>Course objective:</b></p> <ol style="list-style-type: none"> <li>1. Prepare engineering students to analyze cost/revenue data and carry out make economic analyses in the decision making process to justify or reject alternatives/projects on an economic basis.</li> <li>2. Prepare engineering students to function in the business side of professional engineering practice.</li> <li>3. To enhance the decision making capacity of the engineering students by understanding the economics theories</li> </ol>                                                                                                                                                                         |                              |              |                 |
| <p><b>Module-I</b> <span style="float: right;"><b>(10 hours)</b></span><br/> Engineering Economics- Nature, Scope, Basic problems of an economy, Micro Economics and Macro Economics. Demand- Meaning of demand, Demand function, Law of Demand and its exceptions, Determinants of demand, Elasticity of demand &amp; its measurement (Simple numerical problems to be solved), Supply-Meaning of supply, Law of supply and its exception, Determinants of supply, Elasticity of supply, Determination of market equilibrium(Simple numerical problems to be solved).National Income-Definition, Concepts of national income, Method of measuring national income.</p>                             |                              |              |                 |
| <p><b>Module-II</b> <span style="float: right;"><b>(8 hours)</b></span><br/> Production-Production function, Laws of returns: Law of variable proportion, Law of returns to scale, Cost and revenue concepts, Banking -Commercial bank, Functions of commercial bank, Central bank, Functions of Central Bank, An overview of financial system of India.</p>                                                                                                                                                                                                                                                                                                                                        |                              |              |                 |
| <p><b>Module-III</b> <span style="float: right;"><b>(8 hours)</b></span><br/> Basic understanding of different market structures, Determination of equilibrium price under perfect competition (Simple numerical problems to be solved), Break Even Analysis-linear approach (Simple numerical problems to be solved).Inflation-Meaning of inflation, types, causes, measures to control inflation.</p>                                                                                                                                                                                                                                                                                             |                              |              |                 |
| <p><b>Module-IV</b> <span style="float: right;"><b>(10hours)</b></span><br/> Time value of money- Interest - Simple and compound, nominal and effective rate of Interest, Cash flow diagrams, Principles of economic equivalence. Evaluation of engineering projects-Present worth method, Future worth method, Annual worth method, Internal rate of return method, Cost benefit analysis for public Projects, public goods, common property, free rider problem ,market failure and externalities .Depreciation-Depreciation of capital asset, Causes of depreciation, Methods of calculating depreciation (Straight line method, Declining balance method), After tax comparison of project.</p> |                              |              |                 |

PYQs:

2021-22:



Registration no:

2001105213

Subhanshu

Total Number of Pages: 02

B.Tech

Subject Code: HSHM3203

4<sup>th</sup> Semester Regular/Back Paper Examination 2021-22

Subject Name: Engineering Economics

Branch: CSE/ETC/CHEM/PROD/METT

Time: 3 Hours

Max Marks: 60

Answer Question No.1 which is compulsory and any five from the rest.  
The figures in the right-hand margin indicate marks.

Q1 Answer the following questions: (1x10)

- a) Define P/V ratio.
- b) State four factors responsible for demand for a product other than price of a product.
- c) What do you mean by effective interest rate?
- d) Explain breakeven point.
- e) Which organization is responsible for calculating national income in India?
- f) What do you mean by fixed factor?
- g) What do you mean by equal payment series capital recovery amount.
- h) Explain the expansion and contraction in demand curve.
- i) What is time value of equivalence?
- j) Explain how giffin good is an exception of law of demand?

Q2 a) Give rationale behind which stage of law of variable proportion a producer choose to produce? (5)

b) What is law of supply? Explain the factors affecting supply of the product. (5)

Q3 a) Explain the price determination in a perfectly competitive market. (5)

b) Write down the methods of measuring national income. (5)

Q4 a) Determine the effective interest rate for a nominal rate of 6%. (5)

b) Quarterly c) Monthly

What do you mean by Present worth? How this helpful in evaluating the project? (5)

Q5 a) Calculate the present worth of given data if the interest rate is 10%. (5)

| End of year | 0       | 1     | 2     | 3     | 4     | 5     |
|-------------|---------|-------|-------|-------|-------|-------|
| Cash flow   | -10,000 | 2,500 | 3,500 | 4,500 | 5,500 | 6,500 |

a) Assuming interest rate compounded semi annually. (5)

b) Find out future worth.

b) Define depreciation and write down different causes of depreciation. (5)

Q6 a) Explain time value of money. (5)

b) A loan of Rs 500 is made for period of 13 months, from January 1 to January 31 the following year, at a simple interest of 15%. What future amount is due at the end of loan period. (5)

Q7 a) If the quantity demand for chicken increases by 20% when the price of beef increases from Rs.40 to Rs.50, calculate the cross elasticity of demand between chicken and beef. (5)

b) Briefly explain the concept of cost and revenue. (5)

Q8 Write short notes on any two: (5x2)

- a) Annual rate of return.
- b) Isoquant
- c) Constant returns to scale
- d) Difference between straight line method and declining balance method.

Registration no:

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Total Number of Pages: 02

B.Tech.

Subject Code:HSSM3203

4<sup>th</sup> Semester Regular/ Back Paper Examination 2022-23

Subject Name: Engineering Economics

Branch: CSE/ETC/CHEM/METT/PROD. ENGG.

Time: 3 Hours

Max Marks: 60

Question Code:

Answer Question No.1 which is compulsory and any five from the rest.  
The figures in the right hand margin indicate marks.

- Q1 Answer the following questions: (1x10)**
- In case of inferior product income elasticity is \_\_\_\_\_. (Positive/Negative)
  - The first stage of short run production function ends when TP is maximum. (True/False)
  - Which among these two factor is responsible for shifting the supply curve? (Income/Technology)
  - Product differentiation is the characteristics of which market. (Monopoly/Monopolistic)
  - Persistent decrease in price level is known as \_\_\_\_\_. (Inflation/Recession)
  - Total cost = Total fixed cost + \_\_\_\_\_. (Semi variable/Total variable cost)
  - Income received from the sale of output is known as \_\_\_\_\_. (Revenue/Expenses)
  - The expenses on the next best production of firm is known as \_\_\_\_\_. (Opportunity cost/Direct cost)
  - The difference between actual sale and \_\_\_\_\_ is called margin of safety. (Total sale/Break even sale)
  - Overhead is sum of all \_\_\_\_\_ cost of product or services. (Direct/Indirect)
- Q2** a) Define law of demand and law of supply. Using appropriate diagrams, explain why the law of demand does not work in case of precious and inferior product? (5)
- b) If the quantity demand for tea increases by 10% when the price of coffee increases from Rs 0.50 to Rs 0.60, calculate the cross elasticity of demand between tea and coffee. What type of related product the cross elasticity indicates? (5)
- Q3** a) Explain the methods used to measure national income in India. (5)
- b) If there is inequilibrium in a perfectly competitive market then explain how the market functions and equilibrium price and output is restored with a suitable example? (5)
- Q4** a) Write down the concepts of cost and revenue. (5)
- b) Explain the Stage of increasing returns to scale. (5)



- Q5** a) What is depreciation? Why there is need for charging depreciation? (5)
- b) A company has purchased an equipment for Rs 1,00,000 with an estimated life of 8 years. Salvage value of the equipment at the end of estimated life is Rs 20,000. Compute the depreciation and the book value for the year 5. (5)
- Q6** a) Discuss the present worth method of evaluating an engineering alternatives. (5)
- b) The cash flow for an alternative is given below (5)
- | End of year   | 0      | 1    | 2    | 3    |
|---------------|--------|------|------|------|
| Net cash flow | -12000 | 4200 | 6225 | 6330 |
- Evaluate by using present worth method where  $i=10\%$
- Q7** a) Explain the linear Break even analysis with assumptions and limitations. (5)
- b) The fixed cost amount Rs 500000 and the percentage of variable cost to sales is given by  $66 \frac{2}{3}\%$ , if at 100% capacity sales are Rs 300000. Find out the Breakeven point and the percentage of sales when it occurred. Determine the capacity at 80% capacity. (5)
- Q8** Write short notes on any two: (5x2)
- P/V Ratio with its importance.
  - Marginal cost with its importance.
  - Oligopoly.
  - Inflation.

| Registration No. _____                                                                                                                                                                                        |                                                                                                                                                                                                                                                             |               |                           |    |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------|----|----|
|  <b>INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG</b><br>4 <sup>th</sup> Semester Regular/ Back Paper Examination AY 2023-24 |                                                                                                                                                                                                                                                             |               | Total no. of Pages:<br>02 |    |    |
| Branch                                                                                                                                                                                                        | CHEMICAL/CSE/ ETC/ MME/ PRODUCTION                                                                                                                                                                                                                          | Programme     | B. Tech                   |    |    |
| Subject Name                                                                                                                                                                                                  | ENGINEERING ECONOMICS                                                                                                                                                                                                                                       | Semester      | 4 <sup>th</sup> Semester  |    |    |
| Subject Code                                                                                                                                                                                                  | HSHM3203                                                                                                                                                                                                                                                    | Year          | 2023-24                   |    |    |
| Time: 03 Hrs                                                                                                                                                                                                  | Answer Question number 1 which is compulsory and any five from the rest.                                                                                                                                                                                    | Maximum Marks | 60                        |    |    |
| Each Question Carry Marks(M) as Indicated in the right-hand Margin                                                                                                                                            |                                                                                                                                                                                                                                                             |               |                           |    |    |
| Q's                                                                                                                                                                                                           | Questions                                                                                                                                                                                                                                                   | M             | BL                        | CO | PO |
| 1.a                                                                                                                                                                                                           | All of the following are held to be constant when the demand curve of a product is drawn, except for the<br>a) Income of a customer<br>b) Price of the product<br>c) Number of consumers<br>d) Advertisement                                                | 01            | L1                        | 1  | 1  |
| b                                                                                                                                                                                                             | Based on a survey, the income elasticity of demand for Sony television is -0.75. This shows that Sony television is a<br>a) Luxurious product<br>b) Normal product<br>c) Inferior product<br>d) None of the above                                           | 01            | L1                        | 2  | 2  |
| c                                                                                                                                                                                                             | The marginal product curve and average product curve intersect<br>a) Where marginal product equals to zero<br>b) Where total product is at maximum<br>c) At the maximum point of the average curve<br>d) At the maximum point of the marginal product curve | 01            | L1                        | 3  | 3  |
| d                                                                                                                                                                                                             | When output increases in higher proportions than the increases in inputs, the returns to scale is<br>a) Decreasing<br>b) Increasing<br>c) Negative<br>d) Constant                                                                                           | 01            | L1                        | 4  | 2  |
| e                                                                                                                                                                                                             | The cost which changes with the output is called<br>a) Fixed cost<br>b) Variable cost<br>c) Opportunity cost<br>d) Implicit cost                                                                                                                            | 01            | L1                        | 1  | 3  |
| f                                                                                                                                                                                                             | When the total interest is directly proportional to the principal, it is called<br>a) Simple interest<br>b) Compound interest<br>c) Annuity<br>d) None of the above                                                                                         | 01            | L1                        | 2  | 1  |
| g                                                                                                                                                                                                             | Present value is the value today of _____ cash flow.<br>a) Tomorrow's<br>b) Today's<br>c) Future's<br>d) None of the above                                                                                                                                  | 01            | L1                        | 3  | 3  |
| h                                                                                                                                                                                                             | Compound interest of Rs 25000 @12% per annum for two years will be<br>a) 6360<br>b) 6630                                                                                                                                                                    | 01            | L1                        | 4  | 4  |

| c) 6063<br>d) 3063 |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |          |          |       |       |       |               |       |       |      |      |      |    |    |   |   |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|----------|-------|-------|-------|---------------|-------|-------|------|------|------|----|----|---|---|
| i                  | Which method is not used for evaluating project?<br>a) NPV<br>b) ABC<br>c) ARR<br>d) IRR                                                                                                                                                                                                                                                                                                                                                                   |             |          |          |       |       |       |               |       |       |      |      |      |    |    |   |   |
| j                  | In $BC(i) = B + i - C$ , here B denotes<br>a) Equivalent benefit<br>b) Borrowed money<br>c) Barter<br>d) None of the above                                                                                                                                                                                                                                                                                                                                 | 01          | L1       | 2        |       |       |       |               |       |       |      |      |      |    |    |   |   |
| 2.a                | Define the law of demand and law of supply. Using appropriate diagrams, explain how the demand and supply curve illustrates the related laws?                                                                                                                                                                                                                                                                                                              | 05          | L2       | 3        | 2     |       |       |               |       |       |      |      |      |    |    |   |   |
| b                  | Calculate the price elasticity of demand for A. If the price of A increases from Rs 7.00 to Rs 8.00 per kg. and the quantity demand decreases from 70 to 10 kg. that's indicate whether the demand is elastic or inelastic?<br>Calculate the cross elasticity of demand for B when the price of A decreases from Rs 7.50 to Rs 6.50, and quantity demand for B increases from 30 to 70 kg. Find out whether A and B goods are complements and substitutes? | 05          | L3       | 4        | 3     |       |       |               |       |       |      |      |      |    |    |   |   |
| 3.a                | What does production function mean? What are the differences between short run and long run production functions?                                                                                                                                                                                                                                                                                                                                          | 05          | L4       | 1        | 4     |       |       |               |       |       |      |      |      |    |    |   |   |
| b                  | Explain the laws of returns to scale.                                                                                                                                                                                                                                                                                                                                                                                                                      | 05          | L5       | 2        | 2     |       |       |               |       |       |      |      |      |    |    |   |   |
| 4.a                | Explain the methods of measuring national income in India.                                                                                                                                                                                                                                                                                                                                                                                                 | 05          | L5       | 3        | 3     |       |       |               |       |       |      |      |      |    |    |   |   |
| b                  | Following the demand and supply functions of a commodity are given<br>$Q_d = 100 - 20p$<br>$Q_s = -5 + 15p$<br>Find out the equilibrium price and quantity demanded as well supply.<br>Find out Quantity demand and supply, when the price become rs 4, rs 5, and rs 6.                                                                                                                                                                                    | 05          | L4       | 4        | 4     |       |       |               |       |       |      |      |      |    |    |   |   |
| 5.a                | The following figure relates to a manufacturing company.<br><table border="1"> <tr> <th></th> <th>2023(Rs)</th> <th>2024(Rs)</th> </tr> <tr> <td>Sales</td> <td>50000</td> <td>80000</td> </tr> <tr> <td>Profit</td> <td>10000</td> <td>25000</td> </tr> </table> Find out (a) P/V Ratio b) Fixed cost c) Breakeven point d) Margine of safety for 2024.                                                                                                   |             | 2023(Rs) | 2024(Rs) | Sales | 50000 | 80000 | Profit        | 10000 | 25000 | 05   | L3   | 1    | 2  |    |   |   |
|                    | 2023(Rs)                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2024(Rs)    |          |          |       |       |       |               |       |       |      |      |      |    |    |   |   |
| Sales              | 50000                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 80000       |          |          |       |       |       |               |       |       |      |      |      |    |    |   |   |
| Profit             | 10000                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 25000       |          |          |       |       |       |               |       |       |      |      |      |    |    |   |   |
| b                  | Find IRR from the following cash flow.<br><table border="1"> <tr> <th>End of year</th> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tr> <tr> <td>Cash flow(Rs)</td> <td>-4000</td> <td>1000</td> <td>1000</td> <td>1000</td> <td>1000</td> </tr> </table>                                                                                                                                                                                      | End of year | 0        | 1        | 2     | 3     | 4     | Cash flow(Rs) | -4000 | 1000  | 1000 | 1000 | 1000 | 05 | L5 | 2 | 3 |
| End of year        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1           | 2        | 3        | 4     |       |       |               |       |       |      |      |      |    |    |   |   |
| Cash flow(Rs)      | -4000                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1000        | 1000     | 1000     | 1000  |       |       |               |       |       |      |      |      |    |    |   |   |
| 6.a                | In what time will Rs 1500 amounts to Rs 3000 @7% per annum compound interest under rule 72 and under rule 69?                                                                                                                                                                                                                                                                                                                                              | 05          | L4       | 3        | 4     |       |       |               |       |       |      |      |      |    |    |   |   |
| b                  | An investor has an option to purchase a land that will be worth Rs 500000 in six years. If the value of the land increases at 10% each year, how much the investor willing to pay now for the property?                                                                                                                                                                                                                                                    | 05          | L3       | 4        | 2     |       |       |               |       |       |      |      |      |    |    |   |   |
| 7.a                | What is the present worth of a series of 10 years end payments of Rs 1000 each, when the first payment is due today and the interest is 5%?                                                                                                                                                                                                                                                                                                                | 05          | L5       | 1        | 3     |       |       |               |       |       |      |      |      |    |    |   |   |
| b                  | Define depreciation and differentiate between straight line method and declining balance method.                                                                                                                                                                                                                                                                                                                                                           | 05          | L5       | 2        | 4     |       |       |               |       |       |      |      |      |    |    |   |   |
| 8.                 | Answer any two<br>a) Perfect competition vs Monopoly.<br>b) Revenue function.<br>c) Time value of money<br>d) Cost benefit analysis.                                                                                                                                                                                                                                                                                                                       | 05          | L3       | 4        | 2     |       |       |               |       |       |      |      |      |    |    |   |   |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 05          | L2       | 1        | 4     |       |       |               |       |       |      |      |      |    |    |   |   |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 05          | L3       | 2        | 3     |       |       |               |       |       |      |      |      |    |    |   |   |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 05          | L4       | 3        | 4     |       |       |               |       |       |      |      |      |    |    |   |   |

|                                                                                   |                                                                                                                                                                                                                                                                                 |                                                                                                                      |                 |                           |    |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------|---------------------------|----|
|  |                                                                                                                                                                                                                                                                                 | INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG<br>4 <sup>th</sup> Semester Regular/ Back Paper Examination AY 2024-25 |                 | Total no. of<br>Pages: 02 |    |
| Branch                                                                            | CHEMICAL/CSE/ETC/MME/ PRODUCTION ENGG.                                                                                                                                                                                                                                          | Program                                                                                                              | B. Tech         |                           |    |
| Subject Name                                                                      | Engineering Economics                                                                                                                                                                                                                                                           | Semester                                                                                                             | 4 <sup>th</sup> |                           |    |
| Subject Code                                                                      | HSHM3203                                                                                                                                                                                                                                                                        | Year                                                                                                                 | 2024-25         |                           |    |
| Time: 03 Hrs                                                                      | Answer Question number 1 which is compulsory and any five from the rest.                                                                                                                                                                                                        | Maximum Marks                                                                                                        | 60              |                           |    |
| Each Question Carry Marks (M) as Indicated in the right-hand Margin               |                                                                                                                                                                                                                                                                                 |                                                                                                                      |                 |                           |    |
| Q's                                                                               | Questions                                                                                                                                                                                                                                                                       | M                                                                                                                    | BL              | CO                        | PO |
| 1.a                                                                               | ----- goods are goods where the demand is negatively related to income.<br>a) Complement                      b) Normal<br>c) Substitute                        d) Inferior                                                                                                     | 01                                                                                                                   | 1               | 1                         | 1  |
| b                                                                                 | Sellers would offer ----- for sale as price decreases and thus the supply curve is ----- sloping.<br>a) Less, upwards                      b) Less, Downwards<br>c) More, upwards                      d) More, Downwards.                                                      | 01                                                                                                                   | 2               | 2                         | 2  |
| c                                                                                 | Which is the correct indicator of national income?<br>a) GNPmp                              b) NNPfc<br>c) GNPfc                                d) NDPfc                                                                                                                        | 01                                                                                                                   | 2               | 2                         | 3  |
| d                                                                                 | The cost which cannot be recovered once incurred but can be avoided at the beginning is known as<br>a) Opportunity cost                      b) Implicit cost<br>c) Menu cost                              d) Explicit cost                                                     | 01                                                                                                                   | 1               | 2                         | 3  |
| e                                                                                 | Product differentiation is the characteristics of which type of market<br>a) Perfect competition                      b) Monopolistic Competition<br>c) Monopoly                                d) Oligopoly                                                                    | 01                                                                                                                   | 1               | 3                         | 2  |
| f                                                                                 | ----- cost is the additional to the total cost.<br>a) Average cost                              b) Marginal cost<br>c) Total cost                                d) Fixed cost                                                                                                  | 01                                                                                                                   | 1               | 4                         | 3  |
| g                                                                                 | The excess of actual or budgeted sales over the break even sales is known as<br>a) P/V Ratio                                c) Marginal cost<br>b) Contribution per unit                      d) Margin of safety                                                               | 01                                                                                                                   | 1               | 3                         | 3  |
| h                                                                                 | Which of the following is the correct definition of the break even point?<br>The point at which<br>a) Total profit equals total cost<br>b) Total contribution equals total cost<br>c) Total contribution equals total overheads<br>d) Total variable cost equals total revenue. | 01                                                                                                                   | 1               | 4                         | 3  |
| i                                                                                 | The amount can be recovered from the sales of the asset after the expiry of its useful serviceable life.<br>a) Cost of the asset                              c) Salvage value<br>b) Annuity                                      d) Depreciation                               | 01                                                                                                                   | 2               | 3                         | 2  |
| j                                                                                 | Depreciation is charged on<br>a) Variable asset                              b) Depletion<br>c) Fixed asset                                d) Useful life of the asset.                                                                                                         | 01                                                                                                                   | 1               | 4                         | 4  |
| 2.a                                                                               | Define the law of demand and law of supply. Using appropriate diagrams explain how the demand curve and supply curve illustrates the related laws.                                                                                                                              | 05                                                                                                                   | 1               | 4                         | 4  |
| b                                                                                 | If the quantity demand for chicken increases by 20% when the price of egg increases from Rs.0.40 to Rs.0.50, calculate the cross elasticity of demand between chicken and egg.                                                                                                  | 05                                                                                                                   | 1               | 5                         | 2  |

|                       |                                                                                                                                                                                                                                                                                                                                                                       |    |   |   |   |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|---|---|
| 3.a                   | Define the stage of increasing return and diminishing return to stage. Which stage is chosen by the producer for production? Give reason for it.                                                                                                                                                                                                                      | 05 | 3 | 2 | 3 |
| b                     | What is National income? How it is calculated in India?                                                                                                                                                                                                                                                                                                               | 05 | 4 | 3 | 4 |
| 4.a                   | Explain the concept of cost and revenue.                                                                                                                                                                                                                                                                                                                              | 05 | 5 | 5 | 3 |
| b                     | State the difference between perfect competition and Monopoly.                                                                                                                                                                                                                                                                                                        | 05 | 6 | 4 | 4 |
| 5.a                   | State the relationship between Present worth method and annual equivalent worth method.                                                                                                                                                                                                                                                                               | 05 | 4 | 5 | 3 |
| b                     | From the following cash flow stream check the feasibility of the project based on present worth method using $i=10\%$ .<br>End of the year :      0      1      2      3      4      5<br>Cash flow(Rs.) :    -1275    150    300    450    600    750                                                                                                                | 05 | 5 | 4 | 4 |
| 6.a                   | What are the basic assumptions of linear break even analysis?                                                                                                                                                                                                                                                                                                         | 05 | 6 | 5 | 2 |
| b                     | Consider the following data of a company and find the followings<br>P/V Ratio<br>Fixed Cost<br>Break even Sales(Rs.)<br>Year                      Sales(Rs.)                      Profit(Rs.)<br>2024                      10,00000                      200000<br>2025                      15,00000                      400000                                     | p  | 5 | 4 | 3 |
| 7.a                   | Elucidate the fact that "why straight line method is known as Fixed installment method.                                                                                                                                                                                                                                                                               | 05 | 4 | 5 | 4 |
| b                     | Cost of computer Rs.25000<br>Useful life      10 years<br>Salvage value      Rs 2000<br>By using straight line method determine<br>1) Depreciation charge during year 1.<br>2) Depreciation charge during year 2.<br>3) The depreciation reserve accumulated by the end of 3 <sup>rd</sup> year.<br>4) Book value of the computer at the end of 3 <sup>rd</sup> year. | 05 | 3 | 5 | 3 |
| Short Notes (Any Two) |                                                                                                                                                                                                                                                                                                                                                                       |    |   |   |   |
| 8.a                   | Time value of money.                                                                                                                                                                                                                                                                                                                                                  | 05 | 4 | 4 | 3 |
| b                     | Cash flow diagram.                                                                                                                                                                                                                                                                                                                                                    | 05 | 5 | 5 | 4 |
| c                     | Isoquant.                                                                                                                                                                                                                                                                                                                                                             | 05 | 6 | 4 | 4 |
| d                     | Price determination in a perfectly competitive market.                                                                                                                                                                                                                                                                                                                | 05 | 5 | 5 | 4 |